ENRIQUE GARCIA RIVERA

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EDUCATION

Washington University in St. Louis, McKelvey School of Engineering *Expected: May 2027*
B.S. Mechanical Engineering, Minors in Robotics and Mechatronics GPA: 3.64 / 4.00
Relevant Coursework: Modeling, Simulation, and Control (MATLAB/Simulink), Numerical Methods and Matrix Algebra, Fluid Mechanics, Heat Transfer, Solid Mechanics, Vibrations, Electrical and Electronic Circuits for Mechanical Engineers

SKILLS

- **Software and Tools:** MATLAB/Simulink, Python (proficient), C++ (working knowledge), SolidWorks, ANSYS Mechanical, PX4 autopilot, Gazebo simulation, Git/version control, embedded systems (Teensy, Arduino), oscilloscope and multimeter testing.
- **Engineering Competencies:** Sensor instrumentation and fusion, avionics systems integration, telemetry and navigation data analysis, control systems design and simulation, experimental testing and validation.

EXPERIENCE

WashU VTOL – *Avionics & Systems Integration Lead*, St. Louis, MO Sep 2025 – Present
- Designed and integrated avionics architecture for a prototype VTOL aircraft including flight controller, onboard computer, GPS, IMU, telemetry radios, and power distribution systems.
- Configured navigation sensors and telemetry links to support reliable communication between onboard systems and ground monitoring tools.
- Conducted staged subsystem validation through bench testing, collecting and analyzing navigation sensor data to verify sensor functionality, system stability, and data communication.
- Diagnosed and resolved integration issues between sensors, computing hardware, and communication systems, documenting architecture, wiring layouts, and test procedures for repeatable troubleshooting.

Robotics Lab, UMKC – *Research Intern (Grant Funded)*, Kansas City, MO May 2025 – Aug 2025
- Built an embedded sensing platform synchronizing IMU and EMG sensor streams at 1 kHz for robotics experiments.
- Characterized sensor noise, bias drift, and timing jitter across motion capture and arm movement trials to evaluate measurement reliability.
- Collected and prepared IMU and EMG datasets across controlled motion trials to support Extended Kalman Filter development for multi-sensor state estimation.
- Developed Python scripts for signal processing, filtering, and statistical analysis to evaluate filter performance and validate sensor fusion accuracy across motion conditions.
- Presented project results as a research poster at the IEEE Body Sensor Networks Conference (BSN) 2025.

WashU Design Build Fly – *Aerodynamics & Payload Engineer*, St. Louis, MO Sep 2024 – Jan 2026
- Evaluated aircraft wing geometries using aerodynamic analysis tools to study lift and drag performance across design configurations.
- Performed structural finite element analysis in ANSYS to verify wing spar load capacity and structural safety margins.
- Designed mechanical components including payload release mechanisms using CAD modeling and additive manufacturing, documenting procedures and configurations to support repeatable aircraft builds.
- Supported flight testing operations across multiple test sessions, assisting with aircraft setup, observing system behavior, and logging flight data for post-flight review.

Washington University McKelvey School of Engineering – *Course Assistant*, St. Louis, MO Aug 2025 – Dec 2025
- Supported MEMS 2000 Electrical and Electronic Circuits for Mechanical Engineers through laboratory sessions and instrumentation troubleshooting.
- Assisted students with oscilloscope measurements, circuit debugging, and experimental setup.

FedEx Corporation – *Material Handler*, Kansas City, MO Jun 2024 – Aug 2024
- Maintained high throughput while following strict safety procedures in a fast paced logistics environment.
- Demonstrated reliability, attention to detail, and teamwork while operating under time sensitive operational conditions.

TECHNICAL PROJECTS

- **Hyperion Orbital Physics Simulator (C++):** Developed a six degree of freedom orbital dynamics simulator implementing the J2 perturbation model to study spacecraft motion and orbital behavior.
- **Flight Navigation EKF System (Embedded C++):** Implemented a twelve state Extended Kalman Filter combining GPS, barometer, and IMU measurements to estimate navigation states for autonomous systems.
- **Sentinel Prototype (C++ and Python):** Designed and built a robotic sensing system integrating embedded control and sensor feedback for experimental tracking applications.